

Football Analytics AI

Cycle 2 — Expected Goals (xG) Model

Documentation/Findings

1. OVERVIEW

1.1 What Cycle 2 Does

Cycle 2 moves from predicting match outcomes (Cycle 1) to predicting shot quality. The central question changes from 'who will win the match?' to 'what is the probability that this specific shot will result in a goal?' This probability is known in football analytics as Expected Goals, or xG.

For every shot taken in a Premier League match, the model takes a set of pre-shot features — where the shot was taken from, the angle to goal, the distance to goal, the body part used, and the quality of the player — and outputs a single number between 0 and 1 representing the likelihood of a goal.

The cycle is structured across five notebooks: Preprocessing → Exploration → Modelling → Tuning → Explainability.

1.2 The ML Problem Type: Binary Classification

This is a binary classification problem — the model assigns each shot to one of two categories: Goal (1) or No Goal (0). The defining challenge is severe class imbalance: only 10.8% of shots result in goals. The accuracy trap must be avoided — a model that always predicts 'No Goal' achieves 89.18% accuracy while learning nothing.

1.3 Why xG Matters in Football Analytics

- Better than raw goal counts: xG separates luck from genuine quality. A team generating 2.5 xG but scoring 1 goal was unlucky, not poor.
- Tactical insight: Coaches use xG to identify whether shooting positions are high-quality (central, close) or low-quality.
- Player evaluation: Comparing actual goals to cumulative xG reveals elite finishers (above expected) vs underperformers.
- In-game strategy: Real-time xG helps analysts understand the 'true' match state beyond the scoreline.

2. AIM AND OBJECTIVES

Aim

To build a robust, statistically sound Expected Goals (xG) model using only pre-shot features from Premier League event data, evaluated using AUC-ROC to account for severe class imbalance, and explained through SHAP to validate genuine football shooting logic.

Objectives

- Extract and engineer relevant shot features from raw event-level data, removing all post-shot information constituting leakage.
- Perform thorough EDA to quantify class imbalance, understand spatial shot distributions, and measure feature-target correlations.
- Train and compare five models using AUC-ROC as the primary metric.
- Optimise the best-performing model with RandomizedSearchCV, including `scale_pos_weight` in the search space.
- Interpret predictions using SHAP to confirm Distance and Angle dominate in a pattern consistent with football analytics literature.

3. DATASET DESCRIPTION

3.1 Source

Source: `data/raw/events_England.json` — 643,150 total match events from the Wyscout Premier League event dataset, covering all on-ball actions in one season. After filtering to open-play, regulation-time shots only: 8,451 shots from 380 Premier League matches (~22.2 shots/match).

3.2 Target Variable

Value	Meaning	Count	Proportion
1	Goal	914	10.82%
0	No Goal	7,537	89.18%

Class imbalance ratio: 8.25:1. A Dummy model predicting 'No Goal' always achieves 89.18% accuracy — the accuracy trap. AUC-ROC is the only valid primary metric.

3.3 Key Features (9 Features)

Feature	Type	Range	Description
X	Continuous	41–100	Shot horizontal position (% of pitch length; 100=goal line)
Y	Continuous	7–99	Shot lateral position (% of pitch width; 50=centre)

Distance	Continuous	1.25–67.66 m	Euclidean distance from shot position to centre of goal
Angle	Continuous	2.85–180°	Angular width of goal frame from shot position
Left_Foot	Binary	0/1	1 if shot taken with left foot (tag 401)
Right_Foot	Binary	0/1	1 if shot taken with right foot (tag 402)
Header	Binary	0/1	1 if shot was a header or body shot (tag 403)
First_Half	Binary	0/1	1 if shot occurred in the first half
Player_Rank	Continuous	–0.074 to 0.159	Normalised match-specific player quality score

Distance and Angle are derived from (X, Y) coordinates using Euclidean distance and inverse tangent (arctangent) functions respectively. They encode shot difficulty more directly than raw coordinates: shorter distance = goalkeeper has less reaction time; wider angle = more goal exposed.

4. EXPLORATORY DATA ANALYSIS

4.1 The Accuracy Trap

89.18% of shots are non-goals. A model always predicting 'No Goal' achieves 89.18% accuracy — the accuracy trap. The Dummy Classifier gets AUC-ROC = 0.5, correctly exposing it as useless. AUC-ROC measures how well the model distinguishes goals from non-goals at all thresholds, immune to class imbalance effects.

4.2 Spatial Distribution of Goals

Outcome	Mean Distance	Mean Angle
Goals	12.05 metres	38.58 degrees
Non-Goals	19.04 metres	22.57 degrees

Goals cluster tightly near the goal (inside penalty area). Non-goals are spread across the entire attacking half, including many long-range attempts. Distribution is laterally symmetric (no left/right bias), confirming data quality.

4.3 Feature-Specific Findings

Distance (Pearson $r = -0.270$): Strongest negative predictor. Shorter distance $\rightarrow$ higher goal probability. A goalkeeper facing a shot from 5m has far less time and space than one facing a shot from 25m.

Angle (Pearson $r = +0.331$): Strongest positive predictor. Central shots present a wider goal target; byline shots present a narrow sliver. 474 shots (5.6%) flagged as high-angle outliers — investigation shows these are close-range shots (mean 5.4m) with a 41.6% goal rate. DO NOT remove — these are high-quality chances.

Body Part: Headers have 13.52% goal rate vs ~10.2% for foot shots. This is a selection effect — headers are predominantly from close-range crosses, not superior technique. No meaningful handedness effect (left vs right foot).

Match Half: First half 9.81%, second half 11.70% goal rate — ~2% difference. Weakest feature overall.

Player_Rank — 11 $\times$ swing across quartiles:

Quartile	Goal Rate
Q1 (lowest quality)	1.7%
Q2	8.0%
Q3	14.8%
Q4 (highest quality)	19.0%

Despite weak linear correlation (+0.08), Player_Rank has an 11 $\times$ swing from Q1 to Q4 — a strongly non-linear relationship. Tree-based models capture this; Logistic Regression does not.

4.4 Outlier Handling

- Long-range shots (>43m): 20 (0.2%) — legitimate rare attempts. Retained.
- High-angle shots (>51°): 474 (5.6%) — close-range high-quality chances. 41.6% goal rate. DO NOT remove.
- Missing Player_Rank: 85 shots (1.0%) — imputed with median (0.012).

5. PREPROCESSING

5.1 Pipeline

Step	Action	Result
1	Load raw JSON	643,150 events
2	Filter to shot events	Shot events only
3	Remove extra-time shots	Open-play, regulation shots
4	Remove post-shot leakage tags (1201–1218, 1801–1802)	Leakage eliminated
5	Extract shot coordinates (X, Y)	X: 41–100, Y: 7–99
6	Calculate Distance (Euclidean metres)	1.25–67.66 m
7	Calculate Angle (arctangent, degrees)	2.85–180°
8	Create body part binary flags	Left_Foot, Right_Foot, Header columns
9	Create match half binary flag	First_Half column
10	Attach Player_Rank, impute 85 missing with median	Player_Rank complete
11	Create binary target: Goal = 1 if tag 101 present	Goal: 0 or 1
12	Final validation	(8,451, 10): 0 missing
13	Save	wyscout_shots_processed.csv

5.2 Critical Step: Removing Post-Shot Leakage Tags

What are post-shot tags? Wyscout annotates shot events with tags describing where the ball went after the shot: 'goal-low-centre', 'out-high-right', 'blocked-centre', etc. (tags 1201–1218). Tags 1801–1802 are post-event accuracy flags.

Why are these leakage? Both categories are recorded after the shot has been resolved — they describe the result, not the pre-shot conditions. If retained, the model learns 'If the tag says goal-low-centre, it means a goal was scored' — producing near-perfect training accuracy but completely useless for prediction (pre-shot, we have no idea where the ball will go).

This is a subtle leakage type that is easy to miss when working with raw event data for the first time. Its identification and removal is one of the most important contributions of the preprocessing step.

5.3 Class Imbalance: `scale_pos_weight`

`scale_pos_weight` = 8.25 (computed class ratio: 7,537 non-goals ÷ 914 goals). This tells XGBoost to treat each goal example as 8.25× more important than a non-goal during training. For Logistic Regression and Random Forest, the equivalent `class_weight='balanced'` is used.

6. MODELLING

6.1 Train/Test Split

Stratified 80/20 random split (not chronological). Justification: xG is based on physics and geometry — the relationship between shot distance, angle, and goal probability depends on the laws governing ball physics and goalkeeper reach, which do not change meaningfully season to season. A shot from 10m at 35° in 2012 carries the same xG as in 2018. This dataset covers one season only, so there is no multi-season temporal structure to preserve.

- Training: 6,760 shots (731 goals, 10.8%)
- Test: 1,691 shots (183 goals, 10.8%)

6.2 Results Summary

Model	AUC-ROC	Accuracy	Goal Recall	Goal Precision
Dummy Classifier	0.5000	89.18%	0.00	—
Logistic Regression	0.7963	72.50%	0.73	0.24
Random Forest	0.7884	88.82%	0.20	0.46
XGBoost	0.7871	82.32%	0.49	0.30
LightGBM (BEST)	0.8035	77.47%	0.63	0.27

Key observation: Logistic Regression surprisingly outperforms Random Forest and XGBoost (untuned) in AUC. This is because Distance and Angle have strong near-linear signals that a simple linear model can exploit effectively. Random Forest's goal recall (0.20) reveals its voting mechanism is conservative on the minority class despite balanced weighting.

6.3 AUC-ROC Explained

The ROC curve plots True Positive Rate (recall) against False Positive Rate as the decision threshold for predicting 'goal' is swept from 0 to 1. The AUC summarises the entire curve as a single number. Simple intuition: if you pick one shot that became a goal and one that didn't, how often does the model correctly assign higher xG to the goal? An AUC of 0.8183 means it does so 81.83% of the time.

7. HYPERPARAMETER TUNING

7.1 Method

RandomizedSearchCV: 50 combinations × 5-fold Stratified CV = 250 model fits per algorithm. Primary scoring metric: AUC-ROC. `scale_pos_weight` is included in the XGBoost and LightGBM search spaces, allowing the tuning process to verify whether the computed ratio (8.25) is truly optimal.

7.2 Tuning Results

Model	CV AUC	Test AUC	Goal Recall	Key Params
Tuned XGBoost (WINNER)	0.8137	0.8183	0.76	<code>lr=0.01, n_est=200, depth=4, spw=8.25</code>
Tuned Random Forest	0.8120	0.8176	0.77	<code>n_est=300, depth=5, balanced_subsample</code>
Tuned LightGBM	0.8127	0.8152	0.87	<code>lr=0.01, n_est=500, depth=3, spw=12.375</code>

All three models achieve AUC within 0.0031 of each other — extremely competitive. XGBoost selected as winner because: highest AUC (0.8183), strong goal recall (0.76), best probability calibration (confirmed `scale_pos_weight=8.25` optimal), and CV-to-test gap near zero (no overfitting).

Industry context: Published xG models from major football analytics providers typically achieve AUC 0.79–0.84. The Cycle 2 model's 0.8183 is competitive with industry benchmarks.

7.3 Saved Artefacts

- models/cycle2_best_model.pkl — Tuned XGBoost classifier
- models/cycle2_scaler.pkl — StandardScaler (fitted on training data only)

- models/cycle2_feature_cols.pkl — ['X', 'Y', 'Distance', 'Angle', 'Left_Foot', 'Right_Foot', 'Header', 'First_Half', 'Player_Rank']

8. CHRONOLOGICAL VS RANDOM SPLIT

A chronological split is not required for Cycle 2. xG is based on the geometric and physical relationship between shot position and goal probability. The laws governing ball physics, goalkeeper reach, and goal frame dimensions ($7.32 \times 2.44\text{m}$, fixed in all eras) do not change meaningfully from season to season. This dataset covers only one season, providing no multi-season temporal structure to preserve. A stratified random split is therefore both valid and standard for this problem.

Aspect	Cycle 1	Cycle 2
Problem	Match outcome prediction	Shot quality (xG)
Data duration	18 seasons	1 season
Temporal drift	Measurable (−4.17pp leakage effect)	Not applicable
Split strategy	Chronological 80/20	Stratified random 80/20
Reason	Time-series integrity	Geometric relationship is time-invariant

9. EVALUATION

9.1 Confusion Matrix

At the default 50% decision threshold, precision = 0.26 (many false positives) and recall = 0.76 (finds most real goals). The model is far more useful as a continuous probability estimator than as a binary classifier. False positives (predicted goal, actually no goal) are acceptable; false negatives (missed goals) are more costly in xG estimation.

9.2 Strengths and Weaknesses

Strengths:

- Leakage-free pipeline: Post-shot tags 1201–1218 and 1801–1802 removed — AUC 0.8183 reflects genuine predictive ability.
- Industry-benchmark AUC: Competitive with published professional xG models.

- Player skill integration: Player_Rank provides 11× conversion swing — a differentiator from distance/angle-only models.
- Domain-validated explainability: SHAP output precisely matches football analytics understanding.

Weaknesses:

- No shot-creation context: How the chance was created (cross, counter-attack, through-ball) is absent.
- Goalkeeper position unknown: Goalkeeper displacement from centre dramatically affects xG but is unavailable.
- Single season: Training on one season limits exposure to diverse tactical patterns.
- Open-play only: Penalties and direct free kicks require separate models.

10. EXPLAINABILITY (SHAP)

10.1 What SHAP Does in This Context

SHAP is applied to the tuned XGBoost model on the 1,691-shot test set. Baseline xG $\approx$ 0.108 (the training set average goal rate). Each shot's predicted xG is the result of adding all 9 feature SHAP values to this baseline — mathematically consistent and auditable.

Example: Baseline 0.108 + Distance=9m (+0.15) + Angle=45° (+0.09) + Player_Rank=0.12 (+0.04) + Header (−0.02) = Final xG $\approx$ 0.368

10.2 Why Explainability is Critical for xG

An xG model that cannot explain its predictions is dangerous in a practical context. If the model assigns xG = 0.45 to a shot, the analyst must understand why: 'It was 8m from goal at a 50-degree angle by a top-ranked striker.' SHAP provides this reasoning feature by feature. This builds trust, enables debugging, and supports dashboard transparency.

10.3 Global Feature Importance

Rank	Feature	SHAP Interpretation
1	Distance	Dominant — closer shots have dramatically higher xG
2	Angle	Wider angles increase xG; narrow angles decrease it
3–4	X, Y coordinates	Fine-grained spatial information beyond Distance/Angle
5	Player_Rank	Elite players receive higher xG from identical positions

6	Header	Consistently reduces xG vs foot shots in isolation
7–8	Right_Foot, Left_Foot	Small, inconsistent contribution
9	First_Half	Negligible — tightly clustered at SHAP ≈ 0

Validation: The SHAP ranking exactly matches the established football analytics literature on xG — Distance and Angle dominate, Player skill is third. The model has learned genuine football physics, not statistical noise.

10.4 Waterfall Plot

Starting from baseline 0.108, each feature's bar shows how it pushed the predicted xG up or down. For a close-range central shot by an elite player: Distance pushes up strongly (+0.15), Angle pushes up (+0.09), Player_Rank contributes positively (+0.04), Header slightly negative (−0.02). This transforms xG from a black-box number into a transparent, auditable reasoning process shown on the Streamlit dashboard for every prediction.

11. DEPLOYMENT

11.1 Saved Artefacts

- cycle2_best_model.pkl — Tuned XGBoost (AUC 0.8183)
- cycle2_scaler.pkl — StandardScaler (must be applied identically to new data)
- cycle2_feature_cols.pkl — Feature column order enforced at prediction time

11.2 Full Pipeline

```
User clicks pitch position (X, Y)
  → Dashboard computes Distance and Angle
  → Assembles 9-feature vector
  → POST request → FastAPI endpoint
  → Load pkl files → Apply StandardScaler
  → XGBoost.predict_proba(scaled_input)[0][1]
  → Return xG probability
  → Streamlit renders: xG value + pitch heatmap + SHAP waterfall
```

12. CONCLUSION

Cycle 2 delivered a complete Expected Goals model: 643,150 raw events filtered and engineered into a clean 8,451-shot dataset. Severe class imbalance (8.25:1) was handled through scale_pos_weight and AUC-ROC adopted as the primary metric. Five models were compared;

tuning lifted XGBoost to AUC 0.8183 — within industry benchmarks. SHAP confirmed Distance and Angle dominate, matching decades of football analytics research.

13. FUTURE IMPROVEMENTS

- Add shot-creation context: Include how the chance was created (cross, through-ball, counter-attack). Consistently improves AUC by 2–4 points in literature.
- Goalkeeper position estimation: Use player tracking data to include goalkeeper displacement.
- Multi-season training: Extend to 5+ seasons to increase goal examples from 914 to ~4,500+.
- Calibration post-processing: Apply Platt scaling to ensure stated probabilities are empirically accurate.
- Deep learning spatial model: CNN trained on shot heatmaps to capture fine-grained positional patterns.
- Separate model for set pieces: Corners, penalties, and direct free kicks have distinct mechanics.